E50 review pack: Fig. 4g and the text, after the STOP 3 rulings

For the author. Exploratory and post hoc.

2026-11-03

What this is. Fig. 4g as E50 draws it, at final size (87.6 mm wide), with the text and the 4g legend as the author's STOP 3 rulings fix them, the STYLE.md metrics, and the nine Extended Data panels the rulings put in the paper.

- **The rulings:** `docs/2026-11-03_stop3_rulings_consolidated.md` (`cb9b7ce`), committed verbatim and alone.
- **Nothing new is computed.** Every number drawn was stored by E49 (`4fd661b`) and drawn by E50, which was verified (`docs/2026-11-03_e50_result.md`).

The text (rulings section 3).

- **Every sentence is the rulings' own, verbatim, except Methods note N4.** 22 comparisons against the committed file, made each time this pack is built, found no other difference.
- **N4 is aligned to E47 declaration 4a.7, item 5, as section 3 asks.**
  - *As ruled:* The signature comprised **the BCL-2 family members listed in MitoCarta 3.0** whose partial Spearman correlation with OXPHOS at fixed proliferation, averaged over both cohorts, was at least 0.20 in magnitude.
  - *Aligned:* The signature comprised **those of the twelve BH3-only and anti-apoptotic genes listed in MitoCarta 3.0** whose partial Spearman correlation with OXPHOS at fixed proliferation, averaged over both cohorts, was at least 0.20 in magnitude.
  - *Why:* E47 4a.7 applies the rule among the twelve only. Human MitoCarta 3.0 also lists BAX, BAK1, BOK, BCL2L10, BCL2L13 and BNIP3, none of which was a candidate. Read literally, the ruled wording would make the effectors eligible, against N1.
- **The 4g legend is E50's draft with section 4's edits**, items 1 to 5. Item 6 waits on the binding classes.
- **It supersedes E49's result note, Part 4, as committed** (`4fd661b`). The note now carries this text.

The displays (rulings section 1).

- **Accepted as E50 drew them:**
  - the palette from Fig. 1g's source, interpolated in CIELAB;
  - one ±0.6 scale shared by 4g, ED-e and ED-f;
  - the 1.6 mm gap before the Signature row;
  - white ink on 9 cells.
- **The author's, in Illustrator, outside the repo's scripts.** No new display script is needed.
  - **Type sizes:** the group names and the colour bar's title at 6 pt, as in Fig. 1g.
  - **4g's annotations:** the binding code, the signature signs, the § marks and any protein names.
  - **ED-g:** the "Proliferation minus OXPHOS" column is removed.
- **So the panels below are E50's PNG previews, before that work.** The PDF and SVG beside each in `docs/figures/` are the files for Illustrator, with live Helvetica text.
- **Extended Data in the paper:** ED-a, ED-b, ED-e, ED-f, ED-g, ED-h, ED-i, ED-j and ED-k, each cited.
  - **In the repo, out of the paper:** ED-c (`docs/figures/2026-11-03_E48_myc_score_ladder`), superseded by ED-i and ED-j, and S-1 (`2026-11-03_E48_sensitivities_tp53_composition`), kept for referee responses.
  - **ED-d** stays superseded by ED-i.

H2 (rulings section 2).

- **Accepted.** The text says first that the signature was selected for tracking OXPHOS at fixed proliferation.
- **The signature's associations with OXPHOS and proliferation are what the selection intended.** The independent evidence is *BBC3* and *BCL2L1*, MYC activity, and the effectors.
- **The H2 test is not declared.** It is the answer if a referee asks how much of the signature's proliferation coefficient the selection explains.

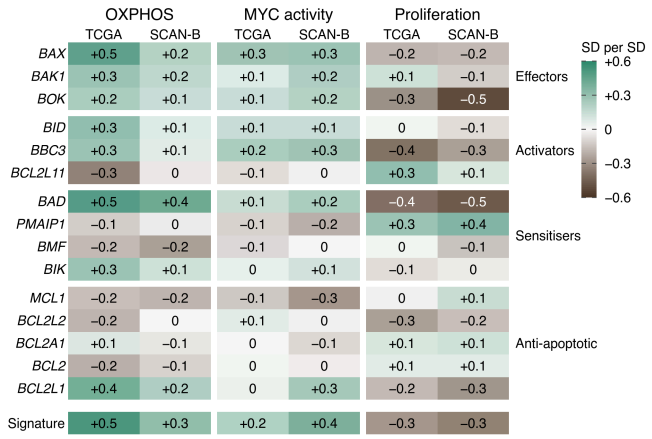
For the author: four observations, not acted on. E49's result note, section 4.1, gives the details of the first three.

1. **R2 gives the model sets, n = 938 and 3,143.** The 2026-09 selection table was computed on all 1,095 and 3,207 tumours (E10's `component_cor`).
2. **R4 names BAD among the genes that rose.** Its explanation is in R7, three sentences later, and brief 5.4 asks for the same sentence or the next. E49's committed text had the same gap.
3. **N4 to N6 restate part of E47 4a.7, item 5,** which stays the Methods' definition. If they replace that paragraph rather than sit beside it, three of its disclosures need to travel:
   - BID's stronger correlation with MYC activity (+0.30) and its split between cohorts (+0.33 and +0.09);
   - that S5's rule uses MYC activity, so S5 is selected against it;
   - the construction sentence, which E47 says may not be dropped.
4. **ED-g's colour scale, ±0.9, was set by the column the author removes.** Without that column, the amendment's rule (symmetric, rounded outward to 0.1) gives ±0.7: the largest remaining value is +0.62 (S7, TCGA, OXPHOS level, reference score).
   - The rulings accept three of E50's open choices and do not mention this one.

- Kept at ±0.9, the remaining cells are paler than the rule would make them. Changing it means redrawing ED-g, and the rulings say no new display script is needed.

How the metrics are counted. Words are whitespace tokens containing a letter or digit. Figure citations in parentheses and reference numbers are not counted.

Fig. 4g (E50)



E50's preview, before the author's Illustrator work: no annotations, and the group names and the bar's title at 5 pt, which become 6 pt.

The text: the rulings' section 3

Sentence ids are for reference. N4 is aligned, as above.

Abstract: the human sentences. [A1] Extending this relationship to 4,081 human breast tumours[8,9], a BCL-2 family signature that tracked OXPHOS at fixed proliferation showed BCL-xL rising more steeply than PUMA. **[A2]** MYC activity, which played no part in defining it, went with the same signature when measured with scores least correlated with proliferation.

Mouse Fig. 4 paragraph: last sentence. [ML] Consequently, established human tumours with high OXPHOS and MYC activity were predicted to carry raised pro-apoptotic signalling restrained by BCL-xL, a configuration distinct from that accompanying proliferation.

Results, paragraph 1: the selection. [R1] In tumours, however, OXPHOS rises with both MYC activity and proliferation (Extended Data Fig. Xa), so testing this prediction required OXPHOS to be separated from proliferation. **[R2]** We therefore searched the twelve BH3-only and anti-apoptotic genes of the BCL-2 family in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143) for those whose expression tracked OXPHOS at fixed proliferation. **[R3]** Six of the eleven mitochondrial members did, and together they formed a signature (Fig. 4g). **[R4]** BCL2L1, encoding BCL-xL, rose with the activators BBC3 and BID and the sensitisers BAD and BIK, whilst MCL1 fell.

Results, paragraph 2: the balance. [R5] Its structure fitted the prediction, BCL2L1 rising more steeply with OXPHOS than BBC3, encoding PUMA, in both cohorts (Extended Data Fig. Xb). **[R6]** This is the reverse of the MYC-expressing gland, in which PUMA rose relative to BCL-xL with OXPHOS priority (Fig. 1i; Extended Data Fig. Xk), and resembles the mouse tumours that sustained high OXPHOS only alongside Bcl-xL (Fig. 3h,i). **[R7]** BAD rose more steeply than BCL2L1 (Fig. 4g), but BAD neutralises BCL-xL only when unphosphorylated, and survival signals phosphorylate it and hold it away from BCL-xL[Zha]. **[R8]** Its expression therefore does not measure its activity, unlike that of PUMA, which is controlled mainly through transcription[Nakano].

Results, paragraph 3: MCL1. [R9] MCL1 fell as BCL-xL rose, so BCL-xL was the only one of the two main guardians to rise alongside PUMA, which binds both[Chen]. **[R10]** MCL1's reported support of respiration in breast cancer stem cells depends on a mitochondrial function separable from its anti-apoptotic role[27]. **[R11]** This suggests that tumours with high OXPHOS rely on BCL-xL rather than MCL1, like persistent OXPHOS-high triple-negative breast cancer cells, which depend on BCL-xL and are insensitive to MCL1 inhibition[26].

Results, paragraph 4: the joint model. [R12] With OXPHOS, MYC activity and proliferation modelled together and intrinsic subtype adjusted for, the signature followed OXPHOS but not proliferation, as its selection intended (Fig. 4g). **[R13]** So did BBC3 and BCL2L1, the two genes named by the mouse before any selection (Extended Data Fig. Xh). **[R14]** MYC activity played no part in the selection, yet it went with the signature in the same direction as OXPHOS (Fig. 4g). **[R15]** This held on the two MYC scores least correlated with proliferation, and was weaker on scores that track proliferation closely (Extended Data Fig. Xa,j). **[R16]** The effectors BAX, BAK1 and BOK, outside the signature, also rose with OXPHOS and MYC activity in both cohorts (Fig. 4g).

The clinical paragraph: the new opening sentence, then its two unchanged sentences. [C1] Having separated OXPHOS from proliferation on PUMA and BCL-xL, we next asked whether the two are also separable in their association with chemotherapy response. **[C2]** Additional adjustment for MYC activity, and then for the mean expression of BCL2L1 and MCL1, did not materially alter the OXPHOS–response association (pooled odds ratios 0.87 and 0.88). **[C3]** The transcriptional coupling of OXPHOS and proliferation therefore concealed a clinically relevant distinction.

The closing paragraph: its human sentences. [Z1] In human breast tumours, BCL-xL rose more steeply than PUMA as OXPHOS rose, and MYC activity went with the OXPHOS-associated signature. **[Z2]** Proliferation, although correlated with OXPHOS, was not part of this configuration and went the other way on chemotherapy response.

Methods notes (the signature's definition is E47 declaration 4a.7, item 5, unchanged and not re-measured here). [N1] The effectors BAX, BAK1 and BOK were fitted for comparison with the groups of Fig. 1g and were not added to the signature. **[N2]** Binding of BCL-xL and MCL1 by each BH3-only protein was taken from published peptide affinities and is shown only where the source states it[Chen]. **[N3]** This analysis was post hoc, and the signature's proliferation coefficient had been seen in an earlier analysis before it was declared; no reading rule was applied, and every coefficient is reported. **[N4]** The signature comprised those of the twelve BH3-only and anti-apoptotic genes listed in MitoCarta 3.0 whose partial Spearman correlation with OXPHOS at fixed proliferation, averaged over both cohorts, was at least 0.20 in magnitude. **[N5]** Each was entered with the sign of that correlation, and MYC activity played no part in the

selection. **[N6]** Variants without BID, which met the threshold by the smallest margin, and with BMF, which met it but is not mitochondrial, gave similar associations (Extended Data Fig. Xg).

The legend: E50′s draft, with the rulings' section 4 edits

256 words; the draft has 177. Associations of OXPHOS level, MYC activity and proliferation with the expression of fifteen BCL-2 family genes, in the groups of Fig. 1g, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each row is one linear model of the gene's expression on OXPHOS, MYC activity and proliferation, with intrinsic subtype, and with tumour purity and leukocyte fraction in TCGA-BRCA; each coefficient is in SD per SD with the other two held fixed, and every variable is rank-normalised within each cohort. Cells give the coefficient, rounded to one decimal; intervals are in Extended Data Fig. Xi. MYC activity is scored on MYC_UP.V1_UP, one of the two MYC activity scores least correlated with proliferation; the reference score is used in Extended Data Fig. Xf, and all four in Extended Data Fig. Xj. Left, binding of BCL-xL and MCL1 by BH3-only proteins, from Chen et al. (2005): filled, binds; open, does not; blank, not yet verified. Signature sign: the direction in which each of the six signature genes enters the signature. § BH3-only proteins whose activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). Bottom, the six-gene signature, selected for its association with OXPHOS at fixed proliferation (Methods). OXPHOS priority is shown in Extended Data Fig. Xe. Outside the signature, BCL2L11, BMF and BCL2 also fell with OXPHOS. At fixed OXPHOS and MYC activity, proliferation went with lower BAX and BOK and higher PMAIP1 and BCL2L11 in both cohorts; its association with BAK1 differed in sign between cohorts, and MCL1 was higher only in SCAN-B.

item	the edit	status
1	adds the sentence on cells and intervals, after the modelling sentence	applied
2	the MYC activity sentence names its place among the four scores, and cites ED-f and ED-j	applied
3	the signature sentence names the bottom row ("Bottom, the six-gene signature")	applied
4	cites ED-e	applied
5	adds the two sentences on patterns outside the text, replacing E49′s proposed additions. The rulings do not place them; they are appended, where those additions stood	applied
6	deletes "; blank, not yet verified" once BID, BIK and BMF are entered	waits on the binding classes (rulings section 5)

STYLE.md does not yet cover legends (its section 9), so only the word count is given.

STYLE.md metrics

paragraph	sentences	words	mean	median	range	under 20	under 12	em-dashes
Abstract	2	47	23.5	23.5	22–25	0 of 2	0	0
Mouse Fig. 4, last sentence	1	27	27.0	27	27–27	0 of 1	0	0
Results 1, selection	4	84	21.0	20.5	13–30	2 of 4	0	0
Results 2, balance	4	90	22.5	21	18–30	2 of 4	0	0
Results 3, MCL1	3	73	24.3	23	20–30	0 of 3	0	0
Results 4, joint model	5	98	19.6	20	15–24	2 of 5	0	0
Clinical paragraph	3	65	21.7	24	13–28	1 of 3	0	0
Closing, human sentences	2	40	20.0	20	19–21	1 of 2	0	0
Methods notes	6	154	25.7	24	18–34	1 of 6	0	0

The four Results paragraphs together: 16 sentences, 345 words, mean 21.6, median 20, range 13–30, 6 of 16 (38%) under 20 words. These are the rulings' section 6 figures, reproduced.

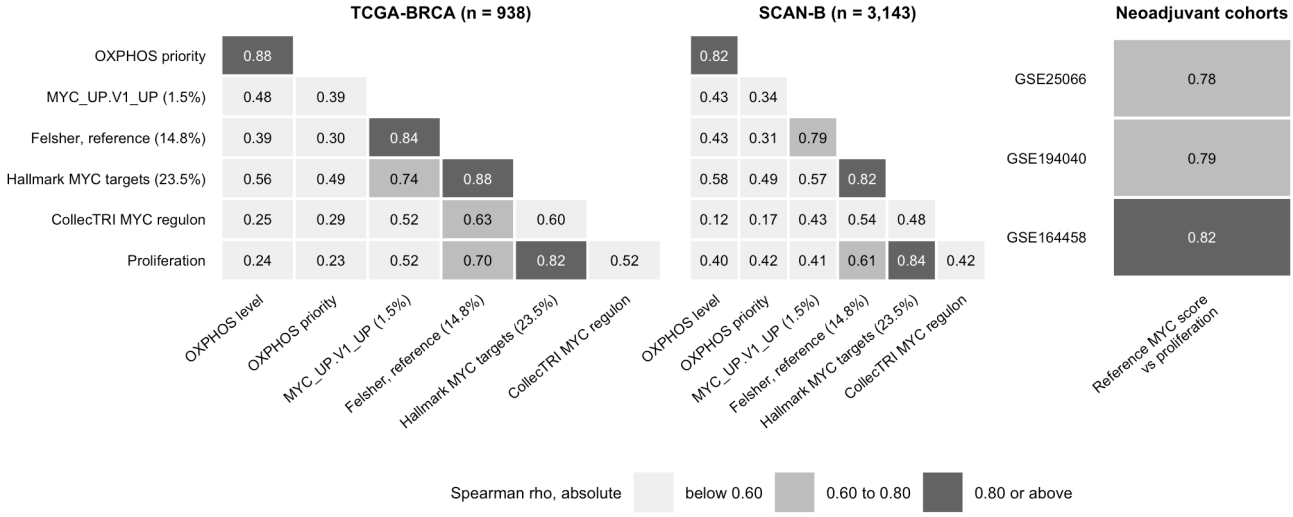
- **Against STYLE.md's targets:** the range and the share under 20 are inside them (12 to 38; 35% to 45%). The mean and the median sit just below them (22 to 25; 21 to 24).
- **No sentence is under 12 words or over 38, and there are no em-dashes.**
- **There are no serial commas.** Each ", and" that the counter flags joins two clauses or two predicates.
- **The new Methods sentences, as ruled, are 30, 18 and 24 words; N4, aligned, is 34.**

The Extended Data panels in the paper

Each is shown at final size, with its legend draft verbatim from the script that drew it. The letters are placeholders: a and b are E48′s; e to h are E50′s, which redraws E49′s; i to k are E49′s. Where each is cited is read from the text and the legend above, and checked against the rulings' 1.4.

ED-a (E48): the scores' correlations

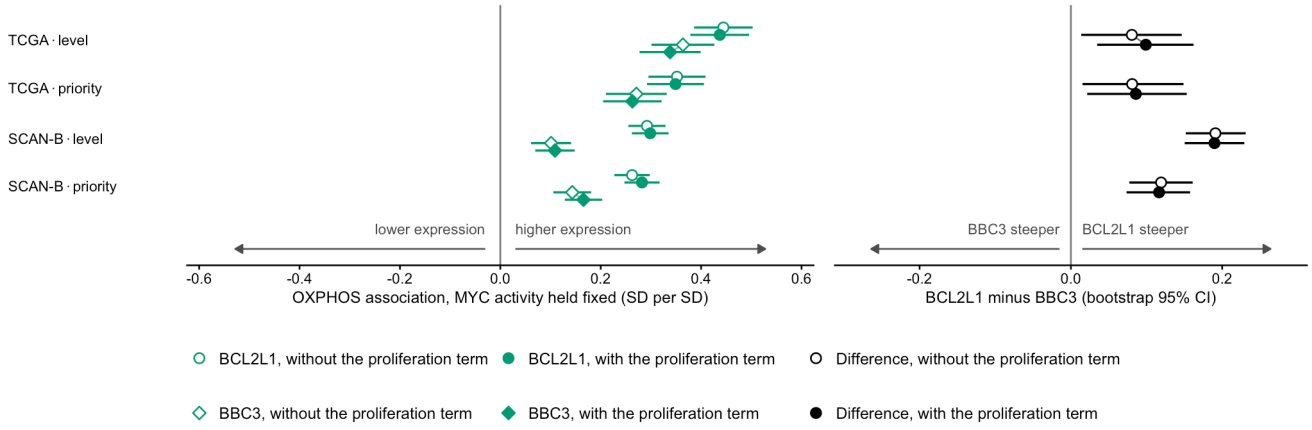
Cited as: Extended Data Fig. Xa, in sentences R1 and R15.



Legend draft (66 words). Spearman correlations among the two OXPHOS scores, four MYC activity scores and proliferation in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), shaded by band (below 0.60; 0.60 to 0.80; 0.80 or above). Each MYC activity score is labelled with the percentage of its genes in proliferation gene sets (Hallmark E2F targets and G2M checkpoint). Right, the reference MYC activity score against proliferation in three neoadjuvant cohorts.

ED-b (E48): *BCL2L1* against *BBC3*

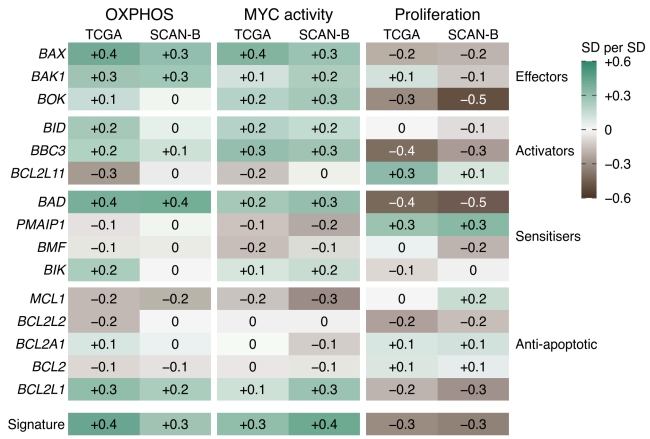
Cited as: Extended Data Fig. Xb, in sentence R5.



Legend draft (51 words). OXPHOS coefficients for BCL2L1 and BBC3 with MYC activity held fixed, without (open) and with (filled) the proliferation term, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), and the difference between them, with 95% percentile intervals from 2,000 paired bootstrap resamples of tumours. Bars on the coefficients, model-based 95% confidence intervals.

ED-e (E50): Fig. 4g on OXPHOS priority

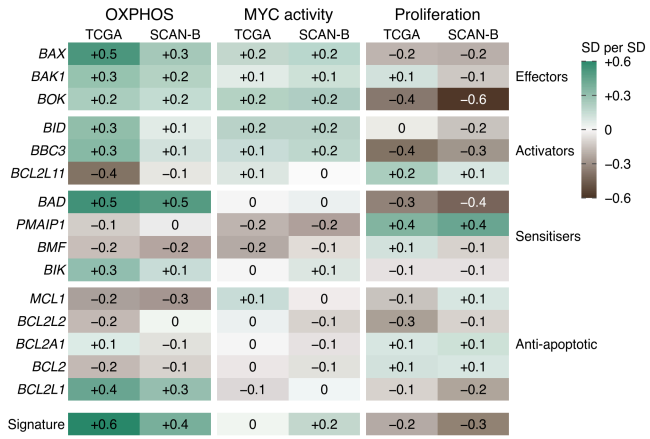
Cited as: Extended Data Fig. Xe, in the 4g legend.



Legend draft (14 words). As Fig. 4g, with OXPHOS scored as its priority within the mitochondrial programme (mitoPPS).

ED-f (E50): Fig. 4g from the reference model

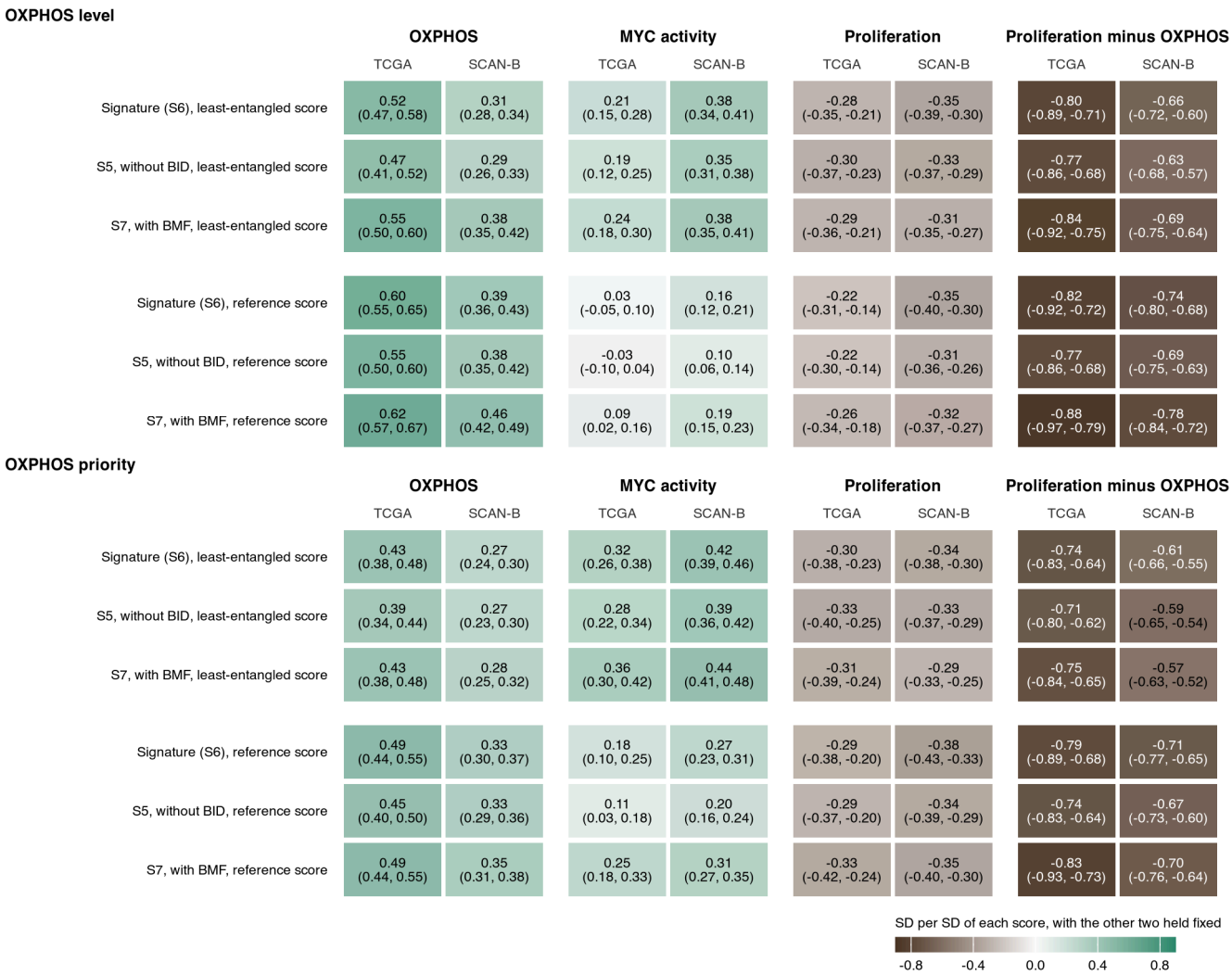
Cited as: Extended Data Fig. Xf, in the 4g legend.



Legend draft (20 words). As Fig. 4g, from models using the reference MYC activity score (Felsher), which correlates with proliferation at 0.61 to 0.70.

ED-g (E50): the signature and its two variants

Cited as: Extended Data Fig. Xg, in Methods note N6.

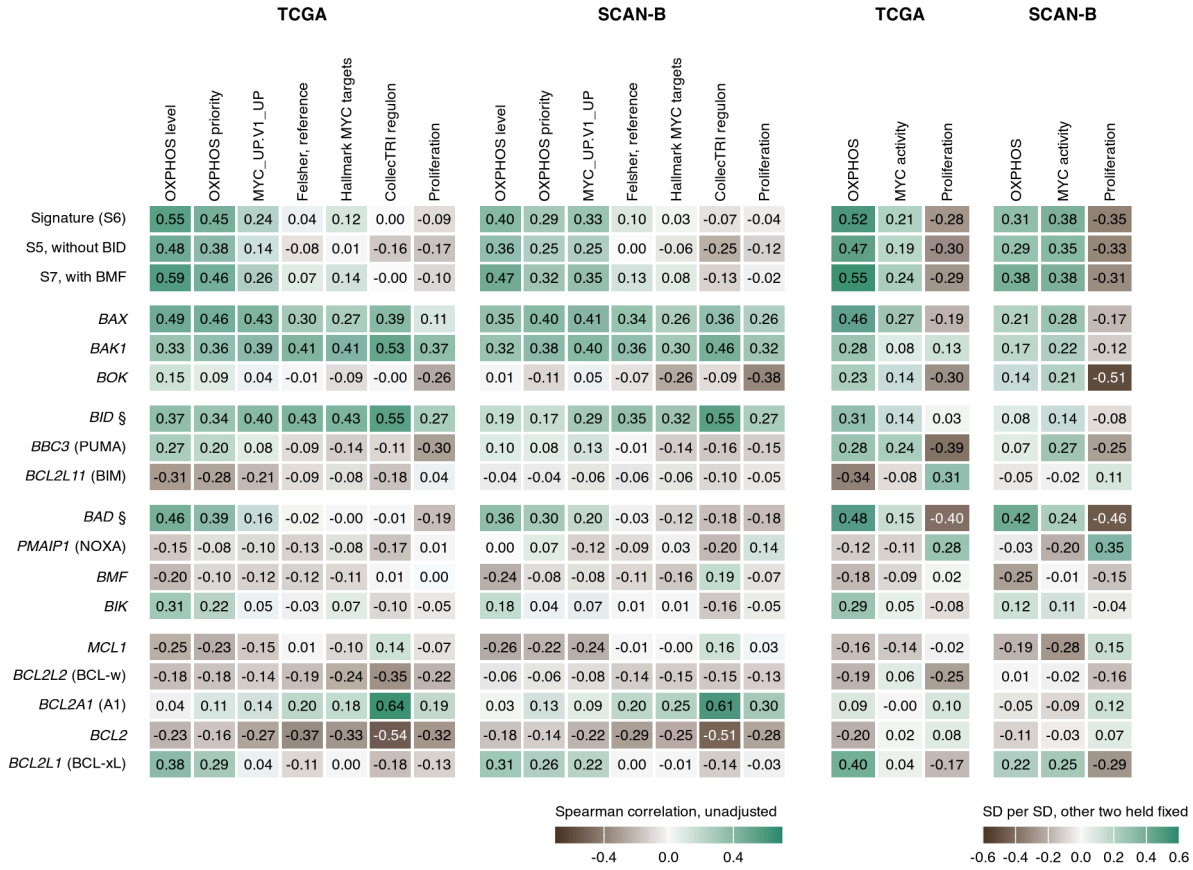


The author removes the "Proliferation minus OXPHOS" column in Illustrator (rulings 1.3). It contains the coefficient that defines the signature, and it remains in E49′s tables. The legend draft does not describe that column, so no legend edit follows.

Legend draft (71 words). The six-gene signature (S6) and its two variants, S5 (without BID) and S7 (with BMF), on the MYC activity score with few proliferation genes and on the reference score, with OXPHOS scored as level and as priority, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each cell gives the coefficient (SD per SD) and its 95% confidence interval. The signature was selected for its association with OXPHOS at fixed proliferation (Methods).

ED-h (E50): raw correlations beside the adjusted coefficients

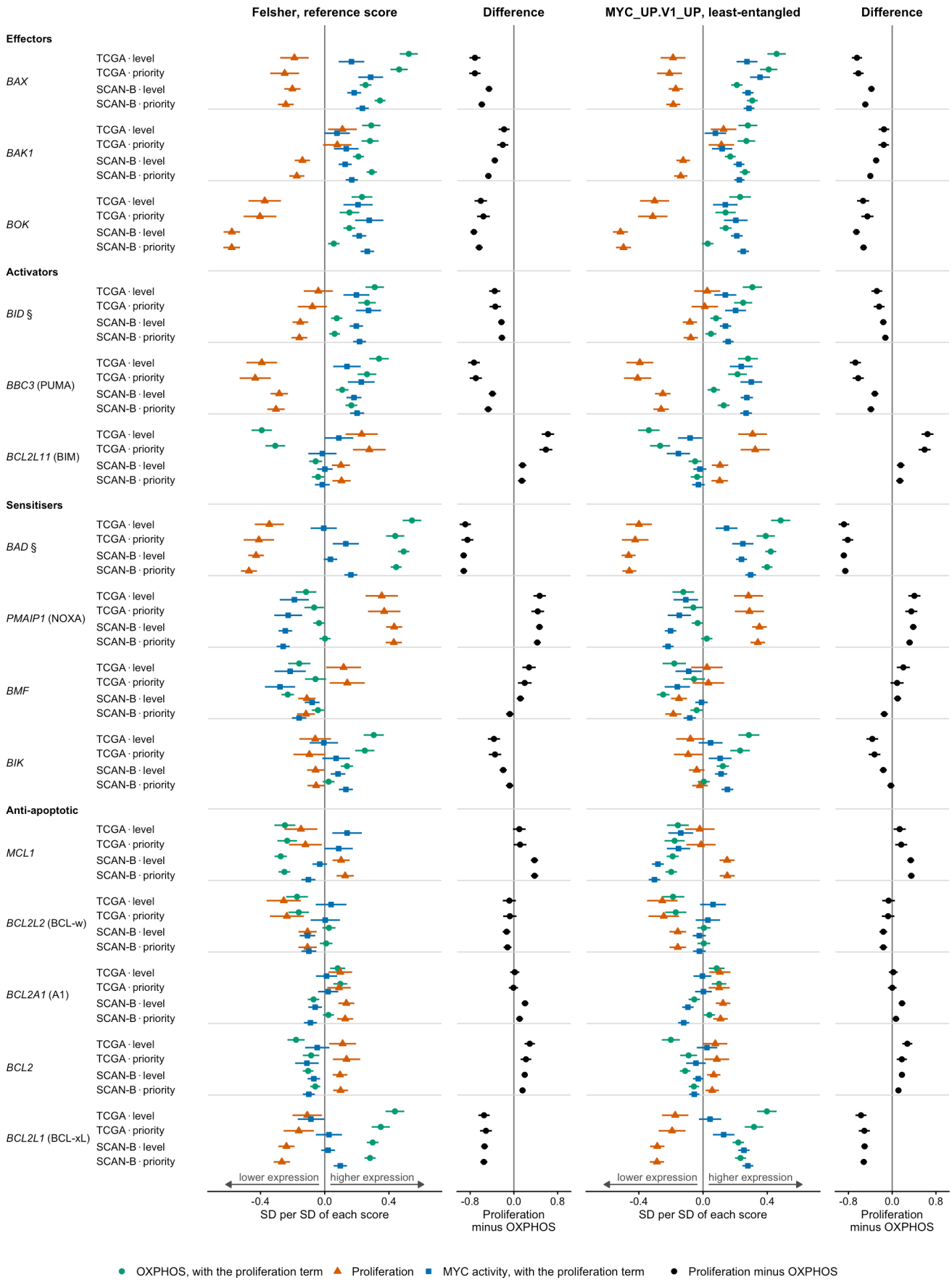
Cited as: Extended Data Fig. Xh, in sentence R13.



Legend draft (53 words). Left, unadjusted Spearman correlations of the signature, its variants and fifteen BCL-2 family genes with OXPHOS (level and priority), four MYC activity scores and proliferation, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Right, the adjusted coefficients of Fig. 4g. The signature was selected for its association with OXPHOS at fixed proliferation (Methods).

ED-i (E49): all fifteen genes, three coefficients and the difference

Cited as: Extended Data Fig. Xi, in the 4g legend.

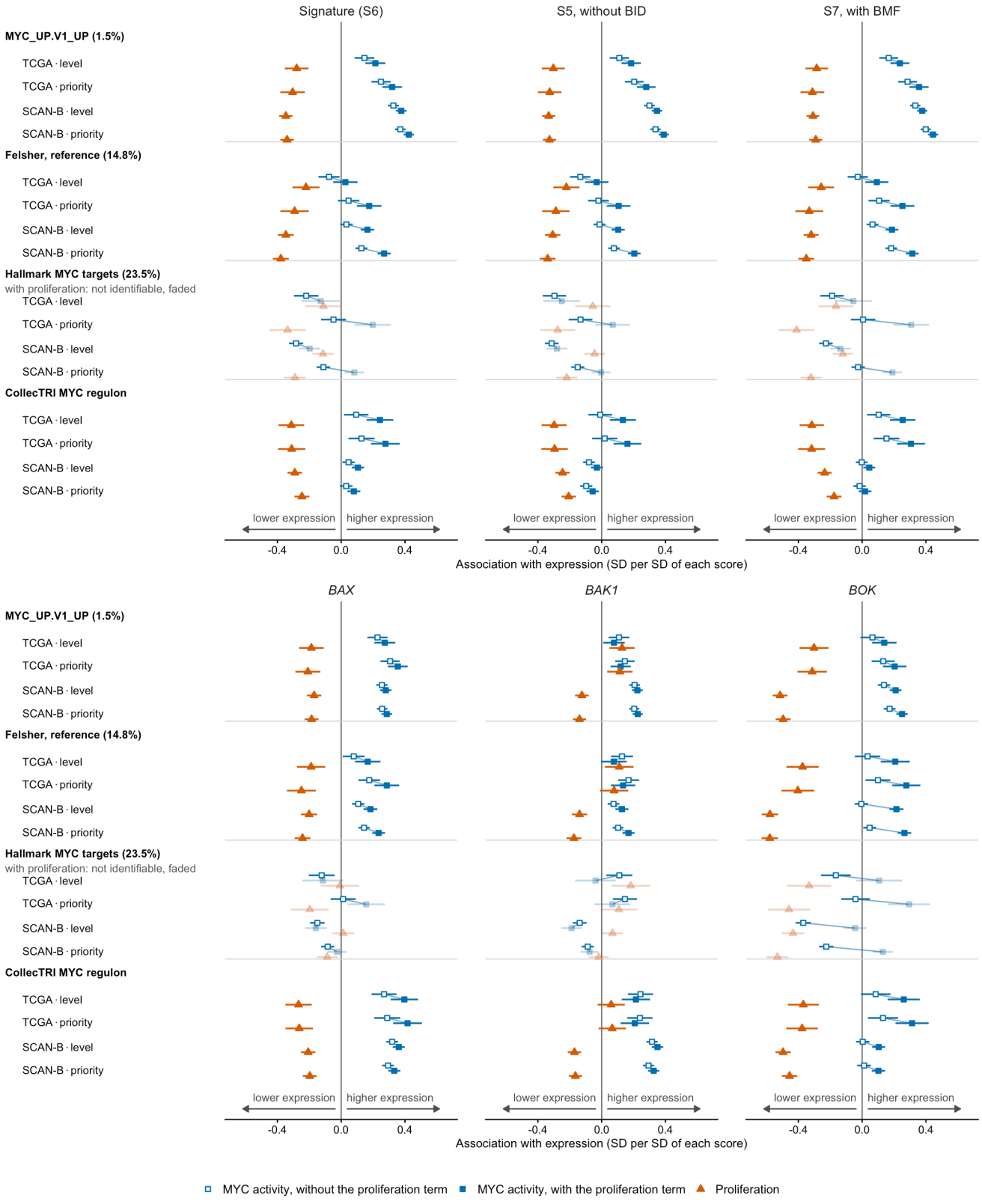


Legend draft (81 words). Fifteen BCL-2 family genes, in the groups of Fig. 1g: coefficients of OXPHOS (circles), proliferation (triangles) and MYC activity (squares), each with the other two held fixed, and proliferation minus OXPHOS, on the reference MYC activity score (left) and the score with few proliferation genes (right), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). § BH3-only proteins whose

activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). No correction for multiple genes. Bars, 95% confidence intervals.

ED-j (E49): MYC activity on four scores, the signatures and the effectors

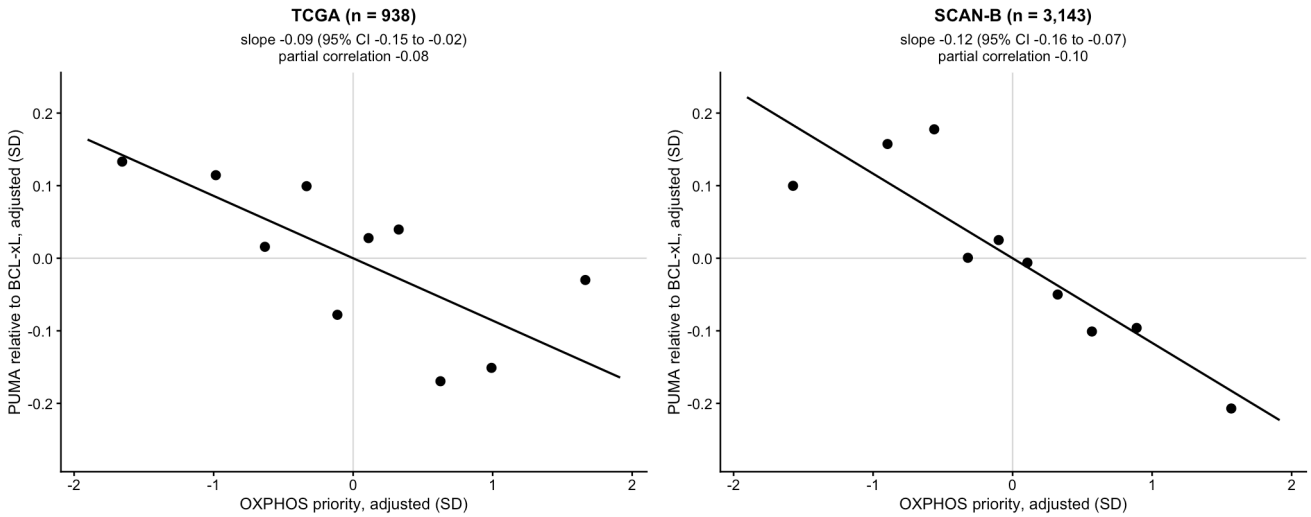
Cited as: Extended Data Fig. Xj, in sentence R15, and in the 4g legend.



Legend draft (76 words). MYC activity coefficients for the signature, its two variants and BAX, BAK1 and BOK on four MYC activity scores, ordered by the percentage of their genes in proliferation gene sets, without (open) and with (filled) the proliferation term, with the proliferation coefficient (triangles), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Faded points are cells in which the score and proliferation correlate at 0.80 or more; they are shown and not interpreted. Bars, 95% confidence intervals.

ED-k (E49): PUMA relative to BCL-xL against OXPHOS priority

Cited as: Extended Data Fig. Xk, in sentence R6.



Legend draft (61 words). BBC3 (PUMA) relative to BCL2L1 (BCL-xL) against OXPHOS priority, each adjusted for the reference MYC activity score, proliferation and intrinsic subtype (and purity and leukocyte fraction in TCGA-BRCA), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Points, means of the ten deciles of adjusted OXPHOS priority; line, the fitted slope, with its 95% interval from 2,000 paired bootstrap resamples of tumours.